

A Comparative Review on Carbon Nanomaterials and Their Impact on Plant Growth With the Lens of Green Chemistry Principles

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Carbon nanomaterials (CNMs), comprising carbon dots, graphene-related materials, and carbon nanotubes, have significant potential for enhancing agricultural productivity. Their compositional compatibility and exceptional properties intrigue a great deal of explorations in agricultural applications, such as fertilizers, pesticides, and regulators of plant growth. However, the evaluation of their agricultural applicability often lacks quantitative sustainability metrics, with insufficient scrutiny on the carbon footprint and scalability of the manufacturing. This review attempts to provide a quantitative ranking system for evaluating the manufacturing processes of the CNMs by applying the twelve principles of Green Chemistry, particularly in the context of agriculture applications. The review also offers a systematically organized account of CNMs' effects on plant systems, encompassing nutrient enhancement, photosynthesis, soil amelioration, disease resistance, and phytotoxicity, which can provide design rationales for the further development of CNMs.

challenge.^[1] In current agriculture practice, inefficient use of fertilizers, pesticides, energy, and water is universal; the waste of them are in the range of 10–75%. The sector is calling for innovative solutions that can revolutionize this traditional industry sector.

Nanobiotechnologies based on carbon nanomaterials (CNMs), synthetic anionic clays, silica nanoparticles, polymers, metal-based nanoparticles, and liposomes have shown promises to step up agricultural productivity and sustainability, owing to the easy uptake and intricate interactions with plants as a result of their nanoscale though.^[2] Nanomaterials for efficient nutrient delivery systems or as vehicles for biomolecule delivery have provided versatile platforms with broad applications to increase agricultural yields.^[3,4] This new-generation

1. Introduction

Sustainable food production is the prerequisite to achieve United Nation's Sustainable Development Goal 2: Zero Hunger. With the rapid growing population, the quest to improve our food production yield for eradicating hunger while significantly reducing the carbon footprint and the consumption of resources is a global

of technologies also focuses on tackling the loss of crop yield due to the exacerbated biotic and abiotic stresses as a result of climate change.^[5,6] On the other hand, the agriculture industry is a relatively low-profit sector, highly sensitive to cost, which puts enormous strains on innovation adoption.^[7] In addition, the ecological impact of agricultural nanof ormulation remains to be understood, which is intrinsically complex.^[7]

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DOI: 10.1002/gch2.202400008

Owing to their exceptional compatibility with the plant system in terms of chemical compositions, as well as their extraordinary electrical, optical, and thermal properties, CNMs are regarded as highly promising candidates in agricultural practices that seek for improved productivity and sustainability.^[8] Although the literature of CNM applications in agriculture is increasing,^[9,10] most of them typically focus on a single type of CNMs regarding their synthesis, and performance in one of the plant health and productivity matrices and in one particular biological environment.^[10,11] It is noted that in previous reports, the sustainability and scalability of the synthesis methods for CNMs have rarely been assessed. The sustainability of synthesis methods of CNMs should consider the carbon cycle and circular economy, as well as pollutant emissions and energy consumption.^[12] Also, it is crucial to make the synthesis process of CNMs scalable, which eventually determines production infrastructure cost, operational costs, product quality consistency, and yields.^[10] Thus, assessments on the sustainability and scalability of the current CNM synthesis methods are a key step for pushing nanotechnology discoveries closer to the market.

On the other hand, insights on CNMs-plant interactions are still lacking.^[13] For instance, the interactions between CNMs and different cytoplasmic organelles, and their influence on various metabolic processes are not elucidated yet. Therefore, the fate, that is, the uptake, translocation, and transformation of CNMs in plants, and the relevant physical barriers associated with the journey of CNMs in plants are of great interest.^[13] In addition, although the effects of CNMs on plants have been reported in a variety of ways, they have not been categorized in a comprehensive manner that reflects the whole life cycle of the nanomaterials from manufacturing to crop application, and to degradation, and that includes comparisons between similar yet starkly different CNMs family members. A number of reviews merely displayed either positive or negative effects of CNMs on plants largely from a biomass-growth point of view without delving into the mechanistic understandings.^[9,11,14] The effects of CNMs on plants can be explicitly classified into micro-nano nutrient delivery, photosynthesis enhancement, soil improvement, and disease resistance improvement, amongst others. A categorical analysis with such a framework can facilitate a more rigorous and systematic understanding of the CNMs-plants interaction for agricultural research.

This review aims to provide a critical assessment on the sustainability and scalability of the synthesis methods of CNMs. Descriptions on the uptake and translocation of CNMs with an emphasis on specific plant organs and their size exclusion limits are also included. The effects of CNMs on plant growth in terms of nutrient improvement, photosynthesis enhancement, soil improvement, disease management, and the relevant mechanisms will be discussed, along with the current critical challenges as a result of climate change, as well as the risk assessment of CNMs in the food chain.^[14–16]

2. Synthesis of CNMs

Recently, green chemistry has received rejuvenated attention because of the looming crisis that the earth is reaching the planetary boundaries. Many synthesis methods and processes used for producing chemicals, which have been taken for granted such as

ammonia synthesis, are under re-evaluation with serious considerations on their impact to the environment.^[17] In addition to sustainability, it is also important to assess synthesis processes in terms of costs, product yields, quality, and process complexity. Hence, the following discussion on synthesis methods focuses mainly on these attributes (Tables 1 and 2).

The synthesis of CNMs can be classified into two approaches (top-down and bottom-up). The top-down approach refers to a process of etching out crystal planes from a precursor, while the bottom-up approach implies that the nanostructures are formed by stacking atoms each other (Figure 1).^[16]

2.1. Carbon Dots

Carbon dots (CDs) have been shown by various studies as a nanoscale toolbox for helping plants cope with biotic and abiotic stresses.^[40] Over the last decade, a number of methods have been developed to fabricate CDs with the desired properties for a particular application. The top-down strategies include chemical exfoliation, chemical oxidation, arc-discharge, and laser ablation while the bottom-up mainly involves pyrolysis/carbonization, microwave synthesis, ultrasonic irradiation, hydrothermal, and solvothermal preparation.

Laser ablation was the first method ever reported for making CDs.^[41] Laser ablation is of short process time and high tunability. Most importantly, among the advantages of laser ablation for CDs is the ability to alter experimental parameters and conditions in favor of smaller particle sizes and the versatility in terms of modification and doping.^[26] However, the equipment for laser ablation is expensive, lacking control for size uniformity, and difficult to scale-up.^[22]

Chemical oxidation is a simple and convenient process that can produce high-quality CDs without expensive devices (Figure 2a). The starting carbon materials are usually cleaved by strong acids or oxidizing agents.^[42,43] These strong acids and oxidizing agents are considered eco-unfriendly, and the process is challenging to scale-up because of the high health and safety risks. Some alternatives using mild oxidants such as hydrogen peroxide have been reported to prepare CDs, but with the hydrothermal-assisted treatment, which could reach 77.3% of the yield.^[44] It is worth noting that there were no by-products in as-prepared CDs aqueous solution and could be directly applied to the application.^[44]

The hydrothermal approach to the preparation of CDs has the advantage of low cost and using the best solvent, water (Figure 2b). Generally, high-pressure and high-temperature hydrothermal reactions are carried out in water solutions enclosed in Teflon vessels in an oven.^[45] Mehta showed a hydrothermal preparation of CDs with a plant-based precursor derived from *Saccharum officinarum* juice.^[46] Hydrothermal synthesis of biomass-derived carbon dots is regarded as green synthesis, which has gained growing interest in recent years.^[47–50] However, questions have been raised on the toxicity of the impurities and byproducts produced in the hydrothermal synthesis.^[50] These impurities might be related to the dye-like substances, giving rise to misconceptions about the nature and characteristics of the CDs. Further studies on cost-effective purification steps are desirable.

Table 1. Comparison of the synthesis method of CNMs in relation to potential industrial commercialization.

Carbon dots (CDs)	Scalability	Cost	Process complexity	Limitations
Chemical oxidation	★	★★	★	Not ecofriendly due to the use of strong acids ^[18]
Laser ablation	★	★★★	★	Poor control over sizes ^[19]
Microwave synthesis	★★★	★	★	Poor control over sizes ^[20,21]
Ultrasonication-assisted	★★	★★	★	High energy cost ^[22]
Hydrothermal	★★★	★★	★	Poor control over sizes ^[23,24]
Solvothermal	★★	★★	★	Poor control over sizes ^[25]
Pyrolysis/Carbonization)	★★★	★★	★	Dye-like substances produced ^[26]
CVD	★	★★★	★★★	High cost ^[27,28]
Carbon nanotubes (CNTs)				
Laser ablation	★	★★★	★★	Method limited to the lab-scale ^[26,29]
Arc discharge	★	★★	★★	Poor control of the structures ^[30]
CVD	★★★	★★	★	Impurities could be introduced ^[31]
Pyrolysis	★	★	★★★	Impurities could be introduced ^[32]
Graphene and graphene-related materials				
Arc discharge	★	★★	★	The number of layers is hard to control ^[33]
CVD	★★	★★★	★★	Impurities could be introduced ^[34]
LPE	★★★	★★	★★★	Low yield ^[35]
SFGP	★★★	★	★★	Lack of parametric and mechanism studies ^[36]
Mechanical exfoliation	★	★★	★★	Different sizes and irregular shapes ^[37]
Oxidative exfoliation-reduction	★★★	★★★	★★★	Batch-to-batch reproducibility ^[38]
Total organic synthesis	★★	★	★	Low dispersibility and occurrence of structural distortion ^[39]

Note: ★-low, ★★★-high.

By replacing water with other organic solvents or ionic liquids in the hydrothermal reactor, method is named solvothermal synthesis.^[25] An eco-friendly method was developed using oxone as a mild oxidant to oxidize natural carbon precursors. This method allows a repeated solvothermal CD production process by recollecting the starting materials, leading to mass production of CDs with high quality.^[51] Low-cost biomass such as wheat and pear juice were also reported to synthesize CDs by the solvothermal method.^[52,53] For example, seaweed was converted into high-quality CDs via oleylamine, which exhibit extraordinary graphitic crystallinity, narrow size distribution (3.6 ± 1.5 nm), and the highest quantum yield (15%) among the CDs synthesized by solvothermal method using crude biomass.^[54]

Pyrolysis has also been frequently used to produce luminous CDs by employing biomass-based carbon precursors. This method has the merits of short reaction times, minimal costs, simple operation, solvent-free, and scalable process. A facile carbonization approach to synthesize CDs from the low-cost sweet red pepper was first reported and obtained CDs that are highly sensitive with down- and up-conversion fluorescence.^[55] The synthesis of CDs from peanut shell waste was developed via a simple and economical carbonization method.^[56] Nevertheless, the duration of pyrolysis is a significant factor to be carefully considered, as dye-like species could be found in the insufficient carbonization process^[57] while some insoluble precipitates were formed under overly sufficient carbonization conditions.^[26] The apparent conflict between the need for sufficient carbonization and the insoluble substances of insufficiently carbonized sam-

ples may represent an intrinsic limitation of the carbonization synthesis method. Therefore, apart from the factor of a period of carbonization, further purification steps to remove the dye-like species and insoluble substances are required, or it is necessary to find how to precisely control the duration of carbonization.

In addition to the strategies above, microwave and ultrasonication-assisted methods offer advantages of greater temperature homogeneity, uniform heating, reduced generation of waste, higher reaction rate, low synthesis time, low operating cost, and high yields.^[58] It is worth noting that the synthesis time could be reduced from several hours to a few minutes.^[59–61] Because microwave technology addresses problems encountered during conventional heating processes, it has become a popular method for nanomaterial synthesis. For instance, materials of poor solubility cause inhomogeneous heating, which then leads to CDs with broad-size distributions.^[62] It was reported that good biocompatibility and high photostability CDs were obtained from *Mangifera indica* leaves by a simple microwave-assisted approach.^[63] Besides, food waste-derived carbon sources by a simple ultrasonic irradiation treatment were shown to have the advantages of good PL properties, low cytotoxicity, and high photostability.^[64]

2.2. Carbon Nanotubes

There are three main methods for carbon nanotubes (CNTs), arc discharge, laser ablation, and chemical vapor deposition (CVD).

Table 2. Green chemistry matrix rating based on given 12 principles from environmental protection agency (EPA).

Green chemistry matrix rating	CDs						CNTs			Graphene			
	Hydrothermal	Microwave Synthesis	Ultrasonication	Solvothermal	Pyrolysis	Chemical oxidation	Arc discharge	Laser ablation	CVD	Arc Discharge	LPE	SFGP	Oxidative exfoliation-reduction
Waste prevention	✓	✓	✓	✓	✓					✓	✓		✓
Atom economy	✓	✓	✓	✓	✓					✓	✓	✓	✓
Less hazardous chemical syntheses	✓	✓	✓	✓	✓					✓	✓	✓	✓
Safer chemicals and products	✓	✓	✓	✓	✓					✓	✓	✓	✓
Safer solvents and auxiliaries	✓	✓	✓		✓					✓	✓		✓
Design for energy efficiency	✓	✓		✓	✓	✓			✓	✓		✓	
Use of renewable feedstocks	✓	✓	✓	✓	✓		✓	✓		✓		✓	
Reduce derivatives		✓	✓										
Use catalysts, not stoichiometric reagents						✓							
Design for Degradation													
Real time analysis for pollution													
Minimize the potential for accidents			✓										
Total rating	★★★★	★★★★	★★★★	★★★★	★★	★	★	★	★	★★★★	★★	★★	★★

Note: two ✓ represents a ★.

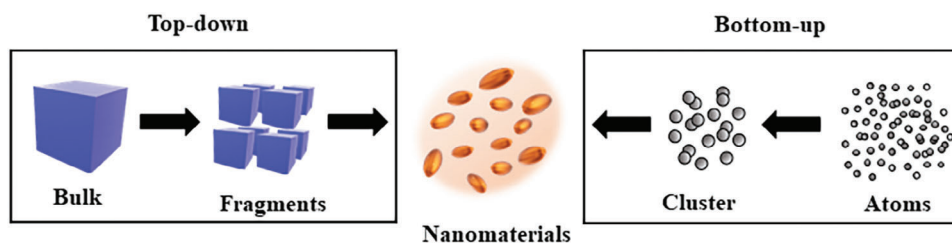


Figure 1. Depicts two commonly used synthesis approaches of CNMs, top-down (left) and bottom-up method (right).

The simplest approach is arc discharge which depends on discharge between graphite electrodes immersed in a chamber with an inert gas. The main advantage of this method is its low cost with a large production of CNTs. However, the quality of CNT is unsatisfactory due to poor control of the structures and impurities when the catalyst is added.^[29] Further purification usually increase the cost leading to impeding its scalability.^[66] An eco-friendly method was developed to produce CNTs by recycling industrial graphite waste and the obtained CNTs have high quality with an average yield, compared with other commercial products.^[67] The use of an aqueous medium in electric arc synthesis contributes to a more environmentally friendly approach compared to traditional methods that involve hazardous chemicals or organic solvents. While the study addresses eco-friendliness to some extent, a more comprehensive environmental assessment, including factors such as carbon emissions and water usage, could provide a clearer picture of the overall environmental impact of the synthesis methods. These factors are also associated with the feasibility of large-scale as it is closely linked to cost-effectiveness.^[67]

Compared to the arc discharge, the main advantage of the laser ablation method is to produce relatively low metallic impurities to CNTs. The drawback of laser ablation is that the obtained CNTs are not uniformly straight and contain some branching (Figure 3a).^[68] In addition, since the procedure requires high-purity graphite rods and high laser power, the laser ablation method is not economically advantageous, and the quantity of CNTs that can be synthesized per day is not as high as the arc discharge technique.^[66,69] Pulsed laser ablation (PLAP) has been

developed as an alternative to traditional continuous-wave laser ablation methods. PLAP allows for precise control over the deposition rate and energy input, resulting in enhanced uniformity and quality of CNTs.^[70] Efforts have been made to optimize the laser parameters, such as pulse duration and energy, to achieve better control over the synthesis process. Additionally, advancements in laser technology and the development of new materials for graphite rods have contributed to reducing the need for high-purity inputs and lowering overall energy requirements. These improvements collectively enhance the efficiency and scalability of CNT production via laser ablation.

CVD is widely considered as the most potential scale-up method.^[30,68] The mass manufacture of CNTs is conducted through the CVD process using transition metal catalysts to break down gaseous hydrocarbon feed into hydrogen and carbon (Figure 3b).^[71,72] Compared with the previous two methods, CVD is rather economically competitive and produces quite pure CNTs with high yields.^[73] Furthermore, Industrial CVD is feasible due to its relative simplicity, process control, energy efficiency, and raw material utilization.^[73] An eco-friendly approach has been developed to synthesize CNTs from non-biodegradable waste tires via CVD. Although this method contributes to waste reduction and recycling efforts owing to the ample and easily accessible nature of waste tires, processing waste tires into a suitable carbon feedstock may involve additional steps, such as pyrolysis or gasification, which could add complexity and energy consumption. These supplementary steps could introduce intricacies and increase energy consumption, thus hindering the scalability of the process. Besides, the strong oxidants (HNO_3 and H_2SO_4) were

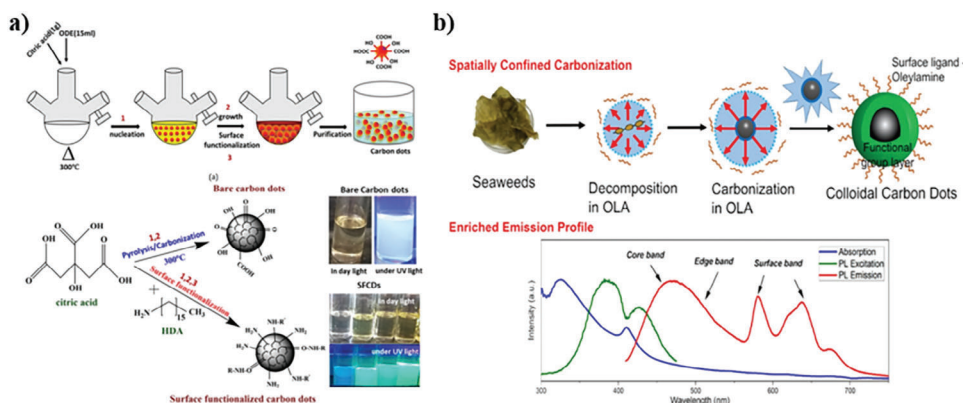


Figure 2. a) Schematic for chemical oxidation synthesis of CDs (reproduced with permission: Copyright 2019, American Chemical Society^[65]) b) The hydrothermal synthesis of CDs (reproduced with permission: Copyright 2023, ACS Publications^[54]).

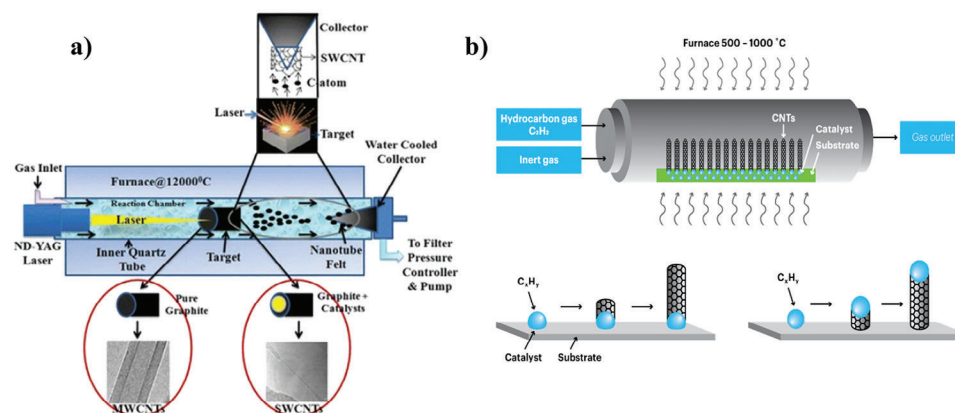


Figure 3. a) Diagram of laser ablation of CNTs (Copyright 2016, Springer Link^[66]) b) Diagram of standard CVD method for the synthesis of CNTs (reproduced with permission: Copyright 2021, Tuball^[76]).

used to purify raw CNTs, which consequently escalating the potential risk associated with effluent disposal^[74]

CNT production from waste plastics by pyrolysis is a relatively new method.^[32] This method has the advantages of resource conservation and environmental protection.^[32,75] However, the complex operation and impurities are the main challenges for further scalability due to the involvement of metal catalysts during production.^[75]

Common purification techniques for CNTs include acid treatment, chromatography, gas-phase oxidation, and liquid-phase oxidation. These techniques have been compared based on the aspects of chemical usage, energy consumption, and the need for specialized equipment in terms of their impact on overall cost. The use of harsh chemicals and sophisticated techniques like chromatography significantly increases costs due to the need for additional chemicals, energy, specialized equipment, and labor. In terms of scalability, many purification techniques present challenges. Techniques like chromatography, although effective at small scales, become cumbersome and less efficient at larger volumes. The requirement for strict control and specialized equipment also hinders the scalability of these methods.

2.3. Graphene and Graphene Related Materials

Two main approaches can be used to synthesize graphene: top-down and bottom-up. The top-down methods usually refer to isolate and delaminate the layers of graphite into single-, bi-, and few-layer graphene, such as mechanical exfoliation, arc discharge, oxidative exfoliation-reduction, liquid-phase exfoliation (LPE). These methods generally provide high-quality products, but low yields with inconsistent products, and are heavily dependent on finite graphite precursors.^[77] The bottom-up method synthesizes graphene from atomic-sized precursors, such as CVD, epitaxial growth, substrate-free gas-phase synthesis (SFGP), template route, and total organic synthesis. The potential industrial production of graphene has been evaluated in terms of cost and environment impacts.^[37]

The arc discharge is a conventional approach to prepare graphene synthesis under inert gas conditions, which is generally expensive due to the use of vacuum equipment (**Figure 4a**).^[78] Wang have reported a relatively low-cost synthesis by using an air medium in the arc discharge, instead of a H_2/He

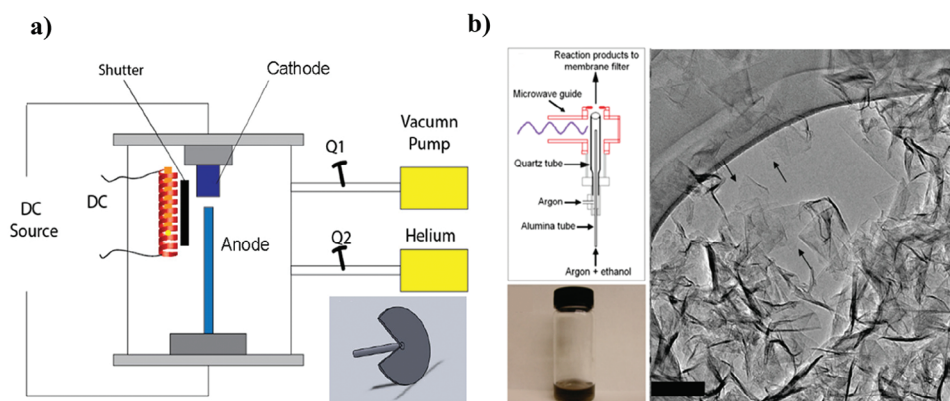


Figure 4. a) Schematic of the set-up of the arc discharge system; and inset represent a shutter used to control the exposure time of the substrate to the carbon flux from the arc (reproduced with permission: Copyright 2015, IOPscience)^[92] b) Substrate-free gas-phase synthesis of graphene sheets (reproduced with permission: Copyright 2008, ACS Publications).^[90]

medium.^[79] By using this method, it is possible to produce several tens of grams of graphene per day and layers ranging between 2 and 10 nanosheets, but high air pressure is required to ensure the growth of graphene. A further effort to reduce the cost of production was found that asphalt is served as electrodes instead of the expensive graphite rods to produce graphene in water. However, the high precision required in process control and control of the number of graphene layers are still the problems of arc discharge, hindering the scalability of graphene production.^[78]

Oxidative exfoliation–reduction is a promising method toward the large-scale production of graphene.^[80] Graphene material produced by this method has been usually utilized as the adsorbent, nanocarrier, and composite reinforcement.^[81–83] A major disadvantage of this method is that it involves reducing agents harmful to humans and the environment. As eco-friendly GO-reducing agents, numerous plant extracts and materials have shown great promise, including tea, eucalyptus bark extract, tannin, carrot, and phytoextracts.^[84,85] However, plant-based reducing agents have some limitations, such as the fact that often they are not as effective as chemical-based reducing agents due to the lower reducing strength, and thus auxiliary agents are usually added.^[85]

LPE is also one of the most commonly used method for graphene production.^[86] For this method, graphite was exfoliated by using suitable solvents followed by purification. Recently, Zhong has reported an eco-friendly solution for liquid phase exfoliation of graphite, which examined three green solvents by their bubble dynamics and acoustic emissions.^[87] Besides, over 40 types of solvents have been studied based on LPE of graphene.^[88] Among these, the less expensive, hazardous solvents were developed for further potential scale-up production, such as cyclopentanone (CPO). Due to its versatility and cost-effectiveness, LPE has a high commercial potential for producing high-quality graphene. Its main advantages, including cheap feedstock, straightforward operations, and environmentally friendly chemicals, make it a viable industrial solution. Process scaling-up, however, faces several challenges, including the lack of information about the commercial reactor setup, high operating costs, and lower yields compared with other synthesis methods.^[89]

Graphene can be synthesized without substrates using a relatively new method known as substrate-free gas-phase synthesis (SFGP) (Figure 4b).^[90] It was reported that this process produces ≈ 2 mg of graphene per minute from 164 mg of ethanol per minute.^[91] Due to its ability to produce clean and high-quality graphene, SFGP has a high potential for industrial scale-up.^[36]

In summary, the current promising synthesis methods of CNMs each have their own advantages and limitations as assessed by the key factors for chemical industrialization including the production cost, reproducibility, processability, and sustainability. Each of these synthesis methods of CNMs has its own niche applications, hence it is difficult to determine a universal approach that addresses all of the criteria unless there is a fundamental breakthrough in its limitations. Under the premise of sustainability with the guidance of green chemistry principles, circularity, waste prevention, and low emission production should have priority in CNMs synthesis and development.

3. Uptake, Transport, and Penetration

It is essential to investigate the fate of CNMs in crops for a better understanding of the CNMs–crops interaction. CNMs' transport routes can be categorized into two length scales where each has different transport barriers: 1) Macroscale, which refers to the movement through plant organ structures such as the leaf cuticle, stomata, and plant vasculature; 2) Microscale, which considers passage through the cell wall.^[93] The transport barriers are crucial to understand the dynamics of CNMs in plants, which further help to evaluate where and how CNMs perform in plants.

Generally, the foliar application and roots treatments are common methods to introduce CNMs to the plants (Figure 5). Plants have a variety of specialized tissues that serve different functions. For instance, compared to root tissue, the structure and function of leaves are substantially different. Similarly, the xylem, or upward transport system, differs significantly from the phloem, namely the downward transport system, in terms of both structure and function. In the following section, we will go through specific plant organs with size exclusion limits and how different tissues of plants affect the journey of CNMs.

3.1. Entry from Leaves and Roots

All leaves have similar features which consist of cuticle, epidermis, stomata, mesophyll, and vascular tissue. The waxy hydrophobic cuticle has a different permeation property with a size exclusion limiting in the nanometer range, which usually impedes the uptake of CNMs but those of a size below 5 nm.^[94] In the cuticle, the CNMs may be transported via lipophilic and hydrophilic pathways. The lipophilic pathway involves solution diffusion within the lipophilic cuticular polymer and hence this route is associated with lipophilicity of CNMs.^[95] On the contrary, hydrophilic pathway usually refers to those aqueous polar pores,^[96] whose diameters are estimated to be ≈ 0.6 to 4.8 nm, and they are likely the entry on cuticle for CNMs.^[97,98] In addition to these nanopores, there are larger pores known as stomata present in the epidermis. Stomata occupy $\approx 5\%$ leaf surface and have two guard cells, forming a pore of ≈ 3 –12 μm wide and 10–30 μm long,^[99] to exchange water and gas with the environment. Therefore, CNMs can easily find their path through the plants when stomata open. Studies have reported that NPs smaller than 100 nm could cross cuticles by disrupting the waxy layer and accumulate in the epidermis where stomata are closed,^[100] whereas particles of 1 μm cannot pass through the cuticle.^[96,101,102] Particles usually get deposited on the cell wall of the substomatal cavity (proximal to the stomata) when crossing the stomata.^[103] Those CNMs that can pass through cuticles will have a better chance to enter plant cells. Furthermore, the structure of the mesophyll also has effects on CNMs' movement due to different plant species. Mesophyll has small intercellular spaces (compact) in xerophytes while large (loose) in some mesophytes and most hydrophytes, which affect the short-distance transport of CNMs after penetrating the epidermis and before entering the vascular system in the leaf.^[104,105]

The CNMs can also be up taken through plant roots. The roots generally consist of the epidermis, cortex, endodermis, pericycle, and vascular system, as illustrated in Figure 5. Among these

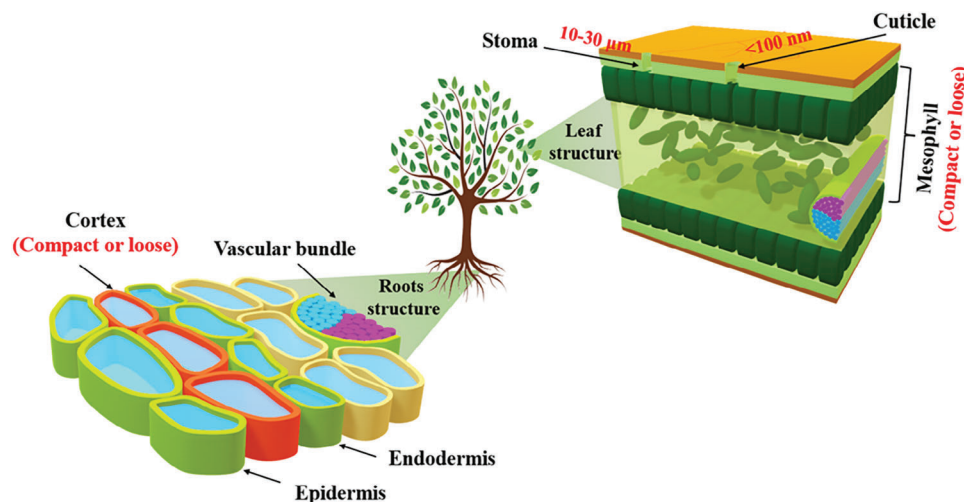


Figure 5. The outer structure of the leaf and roots; and the important plant organs that might affect CNMs journey with the relevant size limits labeled.

structures, the cortex undergoes variation in response to the water content of the soil, which may result in variable levels of resistance to CNM penetration.^[106] The high in water content will cause cortex compaction while the drought condition could lead to large intercellular spaces in the cortex.^[107] Therefore, CNM uptake can be affected by the degree of soil water saturation. A detailed account on interactions between NPs and plant barriers such as root hairs, leaf repellency, pit membrane porosity, xylem segmentation, wounding, nodes, hydathodes, lenticels, trichomes, root mucilage, and exudates can be found in Ref. [13].

3.2. Apoplast and Symplast Pathways

There are two main pathways for NP transport in plants, namely apoplast and symplast (Figure 6). The apoplastic pathway takes

place outside the plasma membrane through extracellular spaces along cell walls, which is essential for the radial movement of CNMs within plant tissues and allows CNMs to further move to the central cylinder of roots (vascular bundle).^[108] Once inside the central cylinder, CNMs can move toward the aerial part of plants through unidirectional xylem. Nevertheless, reaching the xylem requires crossing endodermal cells. The presence of the Casparian strip in endodermal cells will force the movement of CNMs into the symplastic pathway from the apoplastic pathway.^[108] The casparian strip is the last water-impermeable barrier for NPs traveling apoplastically before entering the vascular system, and NPs have been demonstrated to accumulate along the Casparian strip,^[109] as illustrated in Figure 6. The apoplast also serves as a storage to accumulate CNMs because the diameters of longitudinal channels in the cell walls are less than 30 and 40 nm for anticlinal cell walls and epidermal cell walls, respectively.^[13,110]

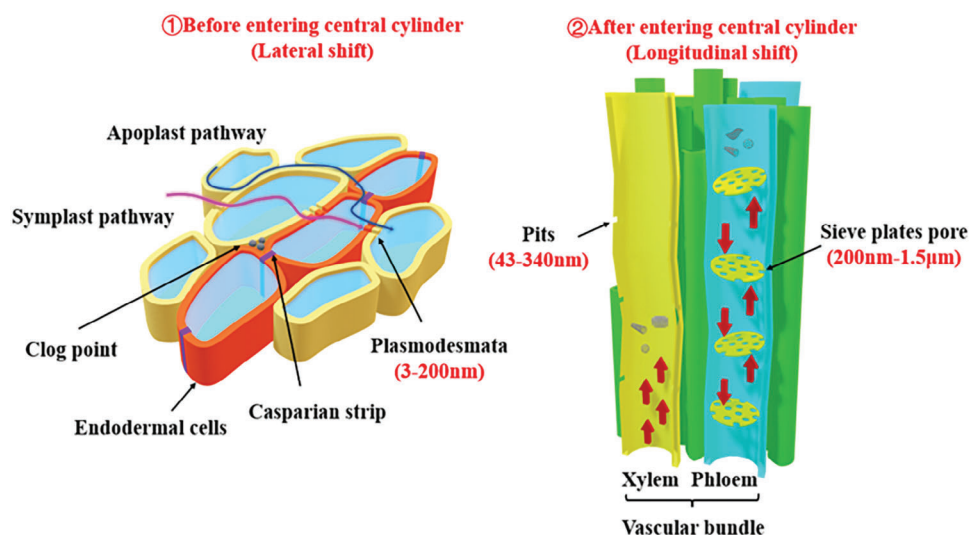


Figure 6. The journey of CNMs in plants. Two pathways before entering vascular bundles have been identified, which are apoplast and symplast. These two ways are responsible for CNMs' lateral shift (left panel). In contrast, after entering the central cylinder, the vascular bundle assists CNM's to move longitudinally so as to reach various sections of plants (right panel).

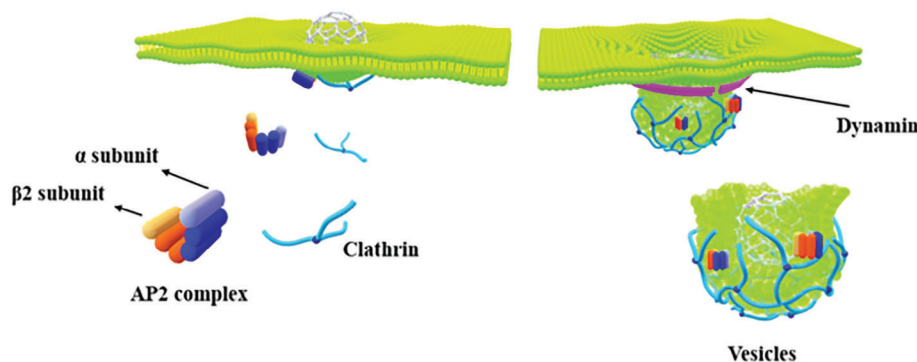


Figure 7. A proposed model of clathrin-mediated endocytosis (CME) in plants.^[114] AP2 complex binds to the plasma membrane to recruit clathrins via subunits protein. The membrane invaginates as more cargos and proteins are recruited and clathrin triskelia polymerize to form the cage. Dynamin assembles at the neck of the bud for scission of the CCV from the PM.

Therefore, the size of CNMs beyond this limit may clog apoplast passage. When in the apoplast, NPs may interact with plant cell walls and membranes, affecting their integrity and permeability. These interactions have been compared based on their effects on plant cell wall and membrane integrity, oxidative stress induction, and disruption of cellular processes in terms of their impact on plant health. Depending on their composition, size, shape, and surface chemistry, NPs may exert toxic effects on plants. Accumulation of NPs in the apoplast can lead to oxidative stress, disruption of cellular processes, and damage to plant tissues. This can result in impaired growth, reduced photosynthesis, and even plant death, thereby highlighting the importance of understanding NP behavior in the apoplast for assessing their overall impact on plant health and function.^[40] On the other hand, the symplastic pathway refers to the movement between the cytoplasm of adjacent cells through plasmodesmata. The plasmodesmata have a size exclusion limit $\approx 3\text{--}4\text{ nm}$ and some molecules larger than the limit such as DNA/RNA, proteins, and transcription factors might have a conformational change in the micro-channels of plasmodesmata.^[13] Due to the strong mechanical strength of CNMs, this means that plasmodesmata are not able to transport $>3\text{--}4\text{ nm}$ CNMs.^[111] However, the plasmodesmata have also been reported to be in the range of $20\text{--}40\text{ nm}$ or $<200\text{ nm}$ in diameter occasionally.^[112] Additionally, the pores of the pits and sieve plates have been reported to be in the range of $43\text{--}340\text{ nm}$ and $200\text{ nm}\text{--}1.5\text{ }\mu\text{m}$.^[113] Therefore, the barriers to CNMs movement such as plasmodesmata, casparian strip, pits, and sieve plate pores are an important influencing factors to affect CNMs' journey in plants. Materials design should be comprehensively considered according to the size of relevant barriers where CNMs will pass by and the physiochemical properties of CNMs.

3.3. Potential Mechanisms of CNMs' Entry into Cell Membrane

Both apoplastic and symplastic pathways eventually require the CNMs to cross the cell membrane before reaching the vascular bundle. There are several mechanisms in the current literature for CNMs to achieve this, although such mechanisms are extensively studied in animal cells and less known in plants.^[114] Endocytosis as a most commonly accepted mechanism provide a gate-

way into the plant cells for foreign materials (**Figure 7**). The endocytosis is classified into clathrin-mediate (CME) and clathrin-independent endocytosis (CIE) in animal cells.^[114] However, only CME has been unequivocally demonstrated in plants and CIE is still poorly defined.^[114] CME starts with the initiation of early invagination at the clathrin-coated plasma membrane, also termed clathrin-coated pits (CCPs), and subsequently forms clathrin-coated vesicles (CCVs) where the CNMs are recognized and packaged for internalization.^[115] Adaptor protein complex 2 (AP2) involved in this process where the subunits of AP2 ($\sigma 2$ and $\mu 2$) bind to the cargo.^[115] The membrane invaginates as more cargo, adaptor proteins, and other accessory proteins are recruited, which a cage-like structure formed by clathrin triskelia polymerizes.^[114] Dynamin assembles at the neck of the vesicle and is responsible for the scission of the CCVs from the plasma membrane in the final stage. Another model regarding CME proposed that a well-defined CCP initiation site pre-exists on the plasma membrane (PM) before the recruitment of AP2.^[115] The size of the CCVs was reported of a size in the range of $70\text{--}120\text{ nm}$ in diameter,^[116] which could be the limiting size of the CNMs that can enter the plant cells. The model of CME assists to further understanding the cell-CNMs interactions at the cellular level. Conjugation of antibodies on CNMs could be developed for addressing molecular motifs such as receptors and cargos to enhance CNMs' functions in cytoplasm, that is, chloroplasts, mitochondria.

Recent research has found CIE mechanism for particle internalization also exists in plant cells. For instance, neutrally charged polyethylene glycol-coated quantum dots in a diameter of 20 nm were sequestered in the cytoplasm of spherical organelle cells ($>120\text{ nm}$) of tobacco.^[117] This indicates that endocytosis without the assistance of clathrin might be a possible mechanism in plant cells. CIE does not entail binding to a specific ligand to a cell membrane and takes place in larger vesicles which is in line with macro-pinocytosis that occurs in animal cells.^[116]

Lipid exchange envelope penetration is another model describing how nanoparticles enter plant cells.^[118] This model has been suggested for explaining interactions between charged NPs and the chloroplast membrane are assumed to soften the membrane, leading to an expanded and fluid state.^[118] In addition, it has been reported that glycerolipids (the predominant lipids) in chloroplast membranes exhibit stronger lipid-nanoparticle interaction

Table 3. Potential interactions between the functional group and specific effect of CNMs on plants.

References	Synthesis methods	Functional groups	Effects
[120]	Electrolysis	—COOH Carbonyl Aromatic skeleton	—OH Ester Ether Enhancement in water uptake and plant growth
[121]	Hydrothermal	—COOH Cyclic ether-containing groups	—OH
[145]	—	—COOH	
[129]	Hydrothermal	—COOH Nitrogen-containing groups	—OH Improvement in photosynthesis and plant growth
[130]	Microwave	Nitrogen-containing groups	
[131]	Hydrothermal	—COOH Nitrogen-containing groups	—OH
[132]	Pyrolysis	—OH Nitrogen-containing groups	
[146]	Chemical Oxidation	—COOH Nitrogen-containing groups	—OH
[147]	Pyrolysis	—OH Nitrogen-containing groups	
[148]	Hydrothermal	—OH Nitrogen-containing groups	
[138]	—	—COOH	Immobilize Cu ²⁺ and reduce its bioaccumulation
[139]	—	—COOH	—OH Cr (VI) reduction
[149]	Hydrothermal	Phenolic hydroxyl <i>Salvia miltiorrhiza</i> molecules/polymers	High antioxidant capacity

than phospholipids (the major constituent) in protoplast membranes.^[119] This might provide an idea that seeking a glycerolipids analog as a cargo attached to CNMs, which helps CNMs be recognized by the cell's membrane, further entering cells.

4. Impacts of CNMs on Plants and Relevant Mechanisms

Though carbon is a vital element for plant growth, the impacts of CNMs on plants are still under exploration. The existing literature reveal seemingly contradictory effects of CNMs between increased plant growth and phytotoxicity, which is calling for further systematic studies to bring clarity before moving into the practical applications. The conflicting impacts primarily stem from the diverse range of plant species. It is challenging to establish an optimal dosage or material configuration due to the unique requirements of different plants or even different plant organs. Certain material designs that are suitable for specific plants may not yield the same results for others. Therefore, it's essential to consider the specific characteristics and needs of each plant type when determining the appropriate dosage and functional group of CNMs. This variability underscores the importance of tailored approaches to CNM application in plant science and agriculture. In this following section, the positive effects of CNMs on plants will be categorized by nutrient benefits, photosynthesis enhancement, soil improvements, and disease resistance.^[40] Also, the phytotoxicity and relevant mechanisms will be discussed with an emphasis on the interaction between functional groups of CNMs and plant growth (Table 3).

4.1. Nutrient Benefits

The nutrient availability could be enhanced by CNMs due to their physiochemical properties. For example, the shoot nutrient contents of tomato at the harvest stage were studied with the treatment of 16 mg kg^{−1} CDs. It is found that the group with the treatment of CDs remarkably increases the contents of shoot nutrients including N, S, P, K, Mg, and Ca.^[120] The relevant mechanism was identified that mineral elements can interact with the functional group of CNMs via hydrogen bonds and electrostatic interactions.^[120] As a result, mineral elements are directly carried to the crops and promote crop growth. This mechanism was also supported by another work, which have recently shown other nutrient changes of lettuce and tomato following treatment with 0.066 mg kg^{−1} CDs.^[121] The contents in V_C and soluble sugar in lettuce plants were significantly enhanced, while there were little differences in the soluble protein content. Meanwhile, the contents of V_C, soluble proteins, and soluble sugars were increased in tomatoes. Besides, similar results were also found in the treatment of CNTs where there was an increase in Ca, Mn, Fe, and P of broccoli.^[122]

In addition, it is well known that water as the medium in plant channels sustaining a variety of compounds for plant development.^[123] It has been reported that aquaporin proteins, namely water channel proteins could be activated by CNMs, thus allowing for more efficient nutrient absorption.^[124–127] For instance, CNTs could significantly activate the expression of water channel protein (*LeAqp2* gene) in tomatoes.^[124] This supported another finding which revealed an upregulation of water channel

protein (*NtPIP1*) associated with the effects of MWCNTs (5–500 mg mL⁻¹) on *Nicotiana tabacum* (*N. tabacum*).^[125] Another recent study also found that CDs could activate the expression of genes encoding aquaporin proteins (*SIAqp2*, *SIPIP2.5*, *SITIP2.1*, and *SITIP2.3*) to accelerate nutrient uptake.^[121]

4.2. Photosynthesis Enhancement

CNMs have been shown to enhance photosynthesis in plants by improving light absorption and energy transfer. This can lead to increased plant growth and productivity. For example, CDs could promote photosynthesis in plants by photoelectric conversion efficiency.^[128] CDs can efficiently absorb light across a broad range of wavelengths, including UV, visible, and near-infrared (NIR) regions. This allows them to capture a larger portion of the solar spectrum compared to traditional photosynthetic pigments. By incorporating carbon dots into the photosynthetic apparatus of plants or algae, light-harvesting efficiency can be improved. CDs can act as light-harvesting antennae, capturing and transferring light energy to photosynthetic reaction centers more efficiently. Further analysis has demonstrated that the CDs can enhance the photosystem activity (Rubisco activity and chlorophyll contents) by increasing the electron transfer rate.^[129] *N. tabacum* leaves that were injected with CDs were found to increase the photosynthetic rate by 18%, enabling the UV light to convert to the blue light used for plants.^[130] Based on that mechanism, a dual-emissive CD was developed, which greatly improves the light-use efficiency in lettuce.^[131] They coated CDs with aromatic clusters and porphyrins to display strong adsorption in the UV range and further emit the blue and red lights that could be absorbed by chloroplasts. Moreover, the recent finding demonstrated near-infrared CDs (NIR-CDs) that can promote photosynthesis by accelerating electron transfer from photosystem II (PS II) to photosystem I (PS I) in *Nicotiana benthamiana* (*N. benthamiana*).^[132] This study also found that NIR-CDs modulated the expression of *PsbP* and *PsiK* genes that are related to photosynthesis and chloroplast synthesis.^[132,133]

4.3. Soil Improvements

Soil is deemed the most crucial habitat for crops, as it is the place where nutrients are stored, water is provided, roots are secured, and all the necessary elements are present for crops to thrive and develop in their growing state.^[134] Thus, the improvement in soil quality will be beneficial to the crop development. CDs have been developed to improve soil quality and fertility under saline-alkali soils, allowing tomatoes to grow better.^[120] Due to the abundant functional groups (carboxyl and hydroxyl), CDs are able to neutralize alkali ions and improve the cations exchange capacity of soil. Since functional CDs have excellent cationic adsorption and desorption properties, saline ions move with water to the subsoil after irrigation, accelerating the leaching of salt and reducing topsoil salinity. Therefore, nutrient cations could fill in the empty soil exchange sites caused by Na⁺ leaching, resulting in higher soil fertility.^[135] Besides, changes in soil physical and chemical properties may promote microbial community. This might increase the input of exogenous carbon as well as the formation

of endogenous carbon, eventually resulting in increased soil organic matter. A rich microbial community leads to more carbon accumulating in the soil through metabolic processes, leading to a positive effect on nitrogen fixation and organic matter degradation, and further providing nutrients with crops.

In soil environments, heavy metals remain a problem. For example, Cr (VI) is highly mobile in soils and toxic to living organisms, whereas Cr (III) is relatively stable and less toxic.^[136] The effect of carbonylated CNTs on Cr(VI) reduction by citric acid was investigated and citric acid was used because it is representative of root exudates and is universally found in soil solutions.^[137] These CNTs could efficiently immobilize Cr (VI) with the adsorption capacity of 8.09 mg g⁻¹, and accelerate the reduction of Cr (VI) by citric acid, which implies their potential use in soil remediation, further enhancing crop growth.^[138] Carboxylated graphene oxide-chitosan spheres (CS) were demonstrated to immobilize Cu²⁺ in soil and reduce its bioaccumulation in wheat plants.^[139] These GO-COOH/CS composites could be used as an efficient adsorbent for Cu²⁺ fixation in soil with a high adsorption capacity of 78 mg g⁻¹. It is worth noting that GO-COOH retained its high performance in Cu²⁺ immobilization as a result of the formation of a spherical structure that prevented soil components from directly contacting adsorptive sites on the GO-COOH/CS. Cu²⁺ bioaccumulation in wheat was significantly decreased by ≈50% using GO-COOH/CS. This research provides a new concept for designing graphene adsorbents for soil remediation.

4.4. Disease Resistance

CNMs can be used to deliver RNA interference (RNAi) molecules to plant cells, which improve plant disease resistance.^[140] RNAi is a process in which small interfering RNAs (siRNA) bind to messenger RNA (mRNA) and prevent the expression of specific genes. In the case of plant disease resistance, RNAi can be used to target pathogen genes that are essential for infection.^[141] When RNAi molecules are combined with carbon nanomaterials, they can be delivered directly to the plant cells, which increases the efficiency and specificity of gene silencing. Besides, the CNMs protect the RNAi molecules from nuclease degradation and provide a stable platform for delivery to the target cells. This leads to a more effective and efficient suppression of pathogen genes and improved plant resistance to diseases.^[141]

Recently, an efficient siRNA delivery by CNTs into intact plant cells through passive transport was demonstrated. This method showed strong silencing of an endogenous *N. benthamiana* gene in plant leaves. With SWNT-mediated siRNA delivery, siRNA is protected from nuclease degradation in cells for up to 24 h and 95% gene silencing efficiency at the mRNA level has been shown. Besides, it was reported that CDs efficiently delivered siRNA into *N. benthamiana* and tomato through a low-pressure spray.^[142] This is the first report that the delivery efficacy of CDs was demonstrated by the silencing of endogenous genes encoding two subunits of magnesium chelatase (an enzyme necessary for chlorophyll synthesis). The strong visible phenotypes were subsequently observed and validated by measuring significant reductions in the target gene transcript and protein levels. These studies have shown CNMs may provide promising tools for

delivering RNA into plants, an essential step for applying RNAi plant biotechnology in agriculture. It should be noted that the use of CNMs for RNAi-mediated plant disease resistance is an emerging area of research, and more studies are needed to fully understand the potential benefits and limitations of this approach.

In addition to RNAi, CNMs have been also shown to enhance the disease resistance of plants on their own surface oxygen-containing CDs were highly effective for increasing the rice plant disease resistance.^[143] The obtained CDs with 29% oxygen content could affect DNA structure and enhance the *Os06g32600* gene expression levels which is related to the plant–pathogen interaction. A positive effect on disease resistance has also been reported with pristine CNTs.^[144] It was found that early foliar exposure of CNTs at 200 mg L^{−1} not only inhibits tobacco mosaic virus (TMV) replication but also limits its subsequent spread to the apical tissues. It is hypothesized that CNTs could enhance the plant's immunity by improving photosynthetic performance and triggering defense responses against TMV, which was reflected in increased levels of antioxidant enzymes and defense-related phytohormones.

Functional groups on CNMs play a crucial role in their functions in plants. Table 3 provides a summary of some examples. It may provide some indications on the potential interactions between the functional group and the specific effect of CNMs on plants. It is found that CNMs with abundant hydrophilic groups (−OH and −COOH) generally increase water uptake and carry more nutrients into the plants. In terms of photosynthesis improvements, it is interesting to observe that the CNMs that have nitrogen-containing functional groups in the examples enhance photosynthesis activities. These nitrogen-containing functional groups are usually of porphyrin rings-like shape.^[131,132]

4.5. Phytotoxicity and Ecotoxicity

Although CNMs have been shown with many positive effects on the plants above, the phytotoxicity of CNMs has been an unavoidable topic that requires rigorous assessments.^[11] The factors of CNMs affecting phytotoxicity mainly are the physicochemical properties of the nanomaterials (concentration/size/categories/stability/surface modification), plant species, the plant growth medium, and the plant growth stage. However, few reports to date have targeted the phytotoxicity of carbon-based nanomaterials,^[150] particularly on the aspect of plant growth medium and plant growth stage. Although some of the studies below are not in the scope of CNMs, CNMs as one of the type of nanoparticles are able to be used for reference from these examples.

The phytotoxicity is a concentration-dependent phenomenon in the plant that varies with different types of carbon nanomaterials. Generally, the CNMs have positive effects on plants within the lower concentration range; while the negative effects within the higher concentration range in most cases. This is because a higher concentration of CNMs might lead to aggregation of CNMs caused by convections, further clogging the vascular system.^[151] It is also crucial to consider the size of CNMs. In carbon toxicology, studies confirmed that short CNTs in size of 30–100 nm transport more quickly due to their size and are less toxic because they could be eradicated easily. In contrast, longer

CNTs could be characterized as toxic because CNTs may enter various organisms, but they could not be quickly degraded in the ecosystem, and thus are more likely to cause toxic effects.^[152,153] However, there are also strong concerns about the toxicology of ultrafine particles, which demonstrated that nanoparticles possessed greater toxicity than those larger size owing to the high reactivity of small materials.^[154] These conflicting viewpoints urgently require more research to elucidate mechanistic pathways and biodistribution of CNMs in plants, providing practical and safe CNM usage. Furthermore, it is important to examine how the physicochemical variables of CNMs influence their effects on plants. The understanding from these will allow the design of CNMs to be more efficient and effective.^[155]

Charges and the charge-related energy surface of CNMs are another parameter to study phytotoxicity. CNMs possess superficial free energy resulting from the exposure of the electrons at the interface of the material and the surround medium.^[156] This high level of superficial free energy supposes an entropy imbalance that tends to diminish spontaneously, through the interaction with other molecules in plant cells or forming aggregates of CNMs. Therefore, such free energy can modify the surface receptors, transporters, specific membrane proteins, and the transport of metabolites and ions to the cell, which further affects phytotoxicity.^[156] NPs with different surface charges have a differential capacity for aggregation and surface properties. In the structure of leaves, the cell wall contains both hydrophobic and hydrophilic components and unequal distribution of fixed negative charges.^[157] Thus, negatively charged cell walls prefer to uptake positively charged nanoparticles in the tissue.

There has also been evidence that the surface functional groups of CNMs influence phototoxicity.^[158] For example, CDs with different functional groups were compared.^[159] The gCDs (synthesized in glycerol) with rich hydroxyl groups showed no toxicity while wCDs (synthesized in water) with higher amounts of amine and carboxyl displayed mortality. Recent studies showing that human myeloperoxidase, eosinophil peroxidase, and soil-relevant fungi-borne laccase can all degrade CDs, have provided some preliminary evidence demonstrating the amenability of CDs to reduce their potential toxicity.^[160,161]

The toxicity investigation on GO with different surface functionalities has also been performed. The results found that unfunctionalized GO induced phytotoxicity in root cells while amine-functionalized graphene oxide was non-toxic and enhanced plant growth.^[162] It is important to recognize that the effects of different functional groups of CNMs on phototoxicity can be complex. A study found that GO that have oxygen- and nitrogen-containing groups magnified the phytotoxicity of arsenic (As) in wheat.^[158] This GO could accelerate the uptake of As, leading to cellular structure damage and electrolyte leakage. Moreover, GO could also promote the transformation of As⁵⁺ to high-toxicity As³⁺. Interactions between CNMs and other contaminants are usually overlooked. It should be noted that when CNMs coexist with pollutants such as heavy metals and organic pollutants their interactions can be either positive or negative, and the surface functional groups appear to hold the key for designing toward lower phytotoxicity.

A critical issue with the design and development of CNMs is constrained comprehension of their ecotoxicity and behavior within environmental systems. It is an urgent task to

systematically assess the potential impact of CNMs on water and soil, whose health can be reflected by a set of abiotic and biotic parameters. In addition to bioassays based on aquatic animals, various in vitro assays using soil and water microbes and algae as well as fish cells have been established as critical indicators for assessing potential ecotoxicity of chemical and biological materials, which can be readily adopted in CNMs research and development.^[163] The work led by Li's group demonstrated for the first time the application of in vitro ecotoxicity assays as the first step assessment on the impact of CDs to photosynthesis and viability of aquatic algae, and soil-borne bacteria.^[54] Overall, ecotoxicity assessment is only at its very early stage for CNM development, more significant R&D efforts are desirable to provide a comprehensive assessment on the pros and cons of CNM applications in crop productions.^[164,165]

5. Conclusion and Future Prospects

In summary, the current promising synthesis methods of CNMs each have their own advantages and limitations as assessed by the key factors for chemical industrialization including the production cost, reproducibility, processability, and sustainability. Each of these synthesis methods of CNMs has its own niche applications, hence it is difficult to determine a universal approach that addresses all the criteria unless there is a fundamental breakthrough in its limitations. Under the premise of sustainability with the guidance of green chemistry principles, circularity, waste prevention, and low emission production should have priority in CNMs synthesis and development. When process scalability is considered, which is essential for real-life applications, environmental protection, carbon emission, simplification, and efficiency should be prioritized.

The twelve principles of Green Chemistry provide a framework for designing or improving materials, products, and processes. Among the three subsets of CNMs for agriculture applications, CDs stand out as the best candidate to implement the twelve principles, though with plenty of room for further improvement. Despite extensive research on CNTs and the availability of many commercial products containing CNTs,^[166] its synthesis methods are far from sustainable compared with other CNMs. There are still problems for CNTs synthesis methods with utilizing strong acids or oxidants as well as fossil fuel precursors such as methane and demanding equipment. The production of GO so far is also reliant either on graphite mining or fossil fuel precursor methane. Moreover, there are three principles that are usually overlooked by the current synthesis methods of all of the three types of CNMs: 1) Design for Degradation; 2) Real Time Analysis for Pollution; 3) Minimize the Potential for Accidents. In terms of designing products, the formulation of chemicals should ensure that they are able to break down into harmless components and do not remain in the environment after reaching the end of their life. Furthermore, facile analytical methodologies need to be developed, which can enable real-time, in situ monitoring and control prior to the formation of hazardous substances or the occurrence of a major incident. This type of analysis can identify changes in process temperature or pH before a reaction gets out of control, which will directly contribute to the safe and efficient operation of chemical plants.

Recent studies on the positive effects of CNMs on plants are promising, such as enhanced germination, accelerated physiological activities and up-regulated gene expression, etc. However, mechanism knowledge concerning the dynamics of CNM interactions with plants and their effect on plant development is still ambiguous. As the data on the CNMs' effects on plants are varied, it is difficult to offer a general conclusion because it depends on many factors such as CNMs' properties (size, shape, charge, concentration, surface modification). In addition, for plant studies, the environmental factors vary significantly in different locations in different seasons, which make the system more complex. An urgent effort is needed to examine how the specific physicochemical variables of CNMs influence their effects on plants. This can be gradually implemented from three perspectives, namely the dynamics of CNMs, physical barriers in crops, and chemical and biochemical reactions between CNMs and plants:

- 1) It is crucial to study the translocation of CNMs. CNMs should be tracked in-real time monitors, since the uptake and shifts of CNMs may indicate the potential location where they reach to, and then we can know what chemical reactions happened in that place and find relevant mechanisms.
- 2) In terms of macroscopic study, more attention should be paid to physical barriers in crops. It is required to understand the crop organs that CNMs might pass through, particularly for their size limit and chemical composition. The organ of crops varies with species. Thus, researchers should be encouraged to identify the common traits of crop organs among various species, which can help develop tailored CNMs to the specific crops, further providing an explicit purpose for designing CNMs.
- 3) At the microscopic level, the mechanistic understanding of the adverse and beneficial effects of the CNMs in crops is still lacking. It is speculated that CNMs may interact or interfere with the crop's metabolic processes. A comprehensive understanding at the level of molecular biology, especially for the interaction between surface functional groups of CNMs and metabolic molecules of crops as well as the chemical transformation, is urgently needed.

Moreover, the inherent versatility of pristine CNMs regarding their optical and electronic properties makes them especially beneficial for agricultural uses. These properties can be further modified by doping carbon-based materials with elements like nitrogen or phosphorous. Doping can improve specific attributes, such as conductivity, chemical reactivity, and mechanical strength, thereby enhancing their effectiveness in various agricultural applications, including sensors, fertilizers, and plant growth regulators.

Acknowledgements

Open access publishing facilitated by Griffith University, as part of the Wiley - Griffith University agreement via the Council of Australian University Librarians.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

carbon dots, carbon nanotubes, graphene, green chemistry principles, phytotoxicity, plants growth response

Received: January 24, 2024

Revised: August 6, 2024

Published online: September 14, 2024

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